Yesterday 8:53 PM

Instance-based
Model-based
explain very easy word and example code concept

In Machine Learning, there are two common approaches: **Instance-based Learning** and **Model-based Learning**.

## 1. Instance-based Learning

**Easy meaning:** The computer remembers old examples and compares new data with them to make predictions.

**Real-life example:** You want to buy a house. The computer checks similar houses that were sold earlier to estimate the price.

Python

```python
from sklearn.neighbors import KNeighborsRegressor

# House areas and prices (in lakhs)
X = [[500], [1000], [1500], [2000]]
y = [20, 40, 60, 80]

model = KNeighborsRegressor(n_neighbors=2)
model.fit(X, y)

print(model.predict([[1200]]))
```

- Algorithm: K-Nearest Neighbors (KNN).
- It compares the new house with similar houses in the stored data.
- It predicts the price using their prices.

## 2. Model-based Learning

**Easy meaning:** The computer studies old data, learns a pattern or mathematical relationship, and uses it to make predictions.

**Real-life example:** The computer learns that house prices generally increase as house area increases.

Python

```
from sklearn.linear_model import LinearRegression

X = [[500], [1000], [1500], [2000]]
y = [20, 40, 60, 80]

model = LinearRegression()
model.fit(X, y)

print(model.predict([[1200]]))
```

- Algorithm: Linear Regression.
- It learns a mathematical relationship between area and price.
- It uses the learned relationship to predict new prices.

## 3. Main Difference

| Instance-based | Model-based |
|---|---|
| Remembers examples | Learns a pattern |
| Compares similar data | Uses a learned model |
| Example: KNN | Example: Linear Regression |
| Prediction may take longer with lots of data | Prediction is often fast |

**Remember this easily:**

- **Instance-based = Remember and compare.**
- **Model-based = Learn and predict.**

Both are types of machine learning, and both can be used to predict real-estate prices.